

Project Report

Plugging into the Future: An Exploration of Electricity Consumption Patterns

Date	06 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
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1. Introduction

1.1 Project Overview

This project analyzes how India's state-wise electricity consumption changed between January 2019 and December 2020, a period spanning the COVID-19 pandemic and nationwide lockdown. Using a dataset covering 33 states across 5 regions (NR, SR, ER, WR, NER), the project builds an interactive Tableau dashboard and guided story to help energy analysts and policymakers explore three scenarios: the overall shift in national consumption trends, regional variation in demand, and the pace of recovery across states after the lockdown.

1.2 Purpose

The purpose of this project is to turn a large, flat spreadsheet of daily state-level consumption records into an accessible, interactive analysis tool. Rather than requiring stakeholders to manually filter and chart the data themselves, the Tableau dashboard and story let them explore national trends, compare regions, and assess lockdown recovery at a glance — supporting faster, better-informed energy planning decisions.

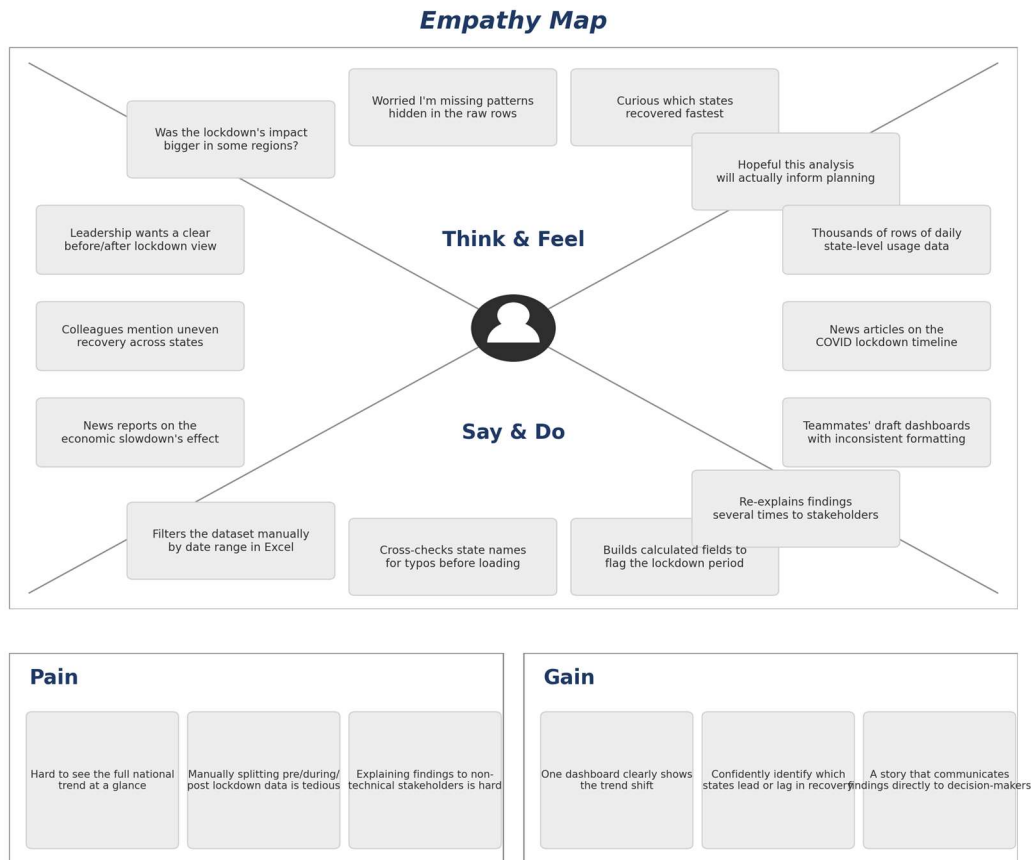
2. Ideation Phase

2.1 Problem Statement

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	An energy policy analyst reviewing India's 2019–2020 state-wise consumption data	understand how national electricity demand shifted before, during, and after the COVID-19 lockdown	the raw data is a flat CSV of thousands of state-date rows with no built-in trend visualization	there is no existing Tableau dashboard summarizing month-by-month national consumption trends	overwhelmed by the sheer volume of unprocessed data
PS-2	A grid planner comparing electricity demand across India's five regions	see how NR, SR, ER, WR, and NER differ in consumption levels and growth	cannot easily compare five regions side-by-side from raw spreadsheet rows	the dataset has no regional aggregation or benchmark view built in	uncertain about where regional attention is most needed
PS-3	A policy researcher assessing how states recovered after the 2020 lockdown	identify which states rebounded quickly and which lagged behind	comparing pre-lockdown, lockdown, and recovery-phase usage per state requires manual filtering across thousands of rows	there is no interactive dashboard segmented by these three time periods	frustrated by how long the manual analysis takes
PS-4	A data analyst preparing a findings report for state electricity boards	build a clear, shareable story explaining consumption trends to non-technical stakeholders	raw numbers and spreadsheets don't communicate the story behind the patterns	there is no narrative-style dashboard connecting the data points into a guided explanation	anxious about whether decision-makers will understand the findings
PS-5	A data analytics student learning Tableau	practice data integration, cleaning, visualization, and	the consumption dataset mixes states, regions, coordinates, and dates	there is no pre-built template for approaching this	challenged translating raw rows into a meaningful visual story

PS	I am	I'm trying to	But	Because	Which makes me feel
	through this capstone project	dashboard/story-building on a real dataset	in one flat file with no guided structure	specific COVID-era energy dataset	

2.2 Empathy Map Canvas



Empathy map for the Energy Policy Analyst persona

2.3 Brainstorming

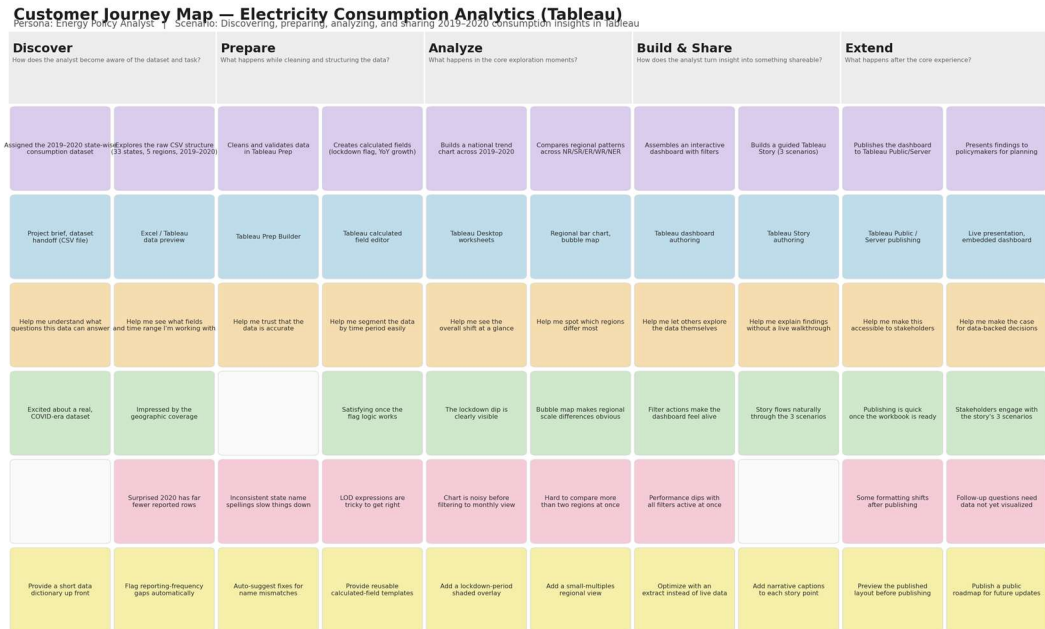
Ideas were brainstormed across six themes — Trend Analysis, Regional Comparison, COVID Recovery Analysis, Visualization Techniques, Dashboard & Story, and Data Prep & Integration — then prioritized by importance and feasibility:



Idea prioritization: importance vs. feasibility

3. Requirement Analysis

3.1 Customer Journey Map



Journey of an Energy Policy Analyst: Discover → Prepare → Analyze → Build & Share → Extend

3.2 Solution Requirement

Functional Requirements:

FR No.	Requirement (Epic)	Sub Requirement
FR-1	Data Integration	Connect Tableau to the state-wise consumption CSV source; refresh into a Tableau extract (.hyper)
FR-2	Data Preparation	Clean/deduplicate records; standardize state names; create calculated fields (lockdown flag, Top N rank)
FR-3	Data Visualization	Trend/bar/pie charts by region; state consumption maps; year-over-year comparison
FR-4	Dashboard	Interactive filters (state, region, year); Top N / Bottom N ranking; combined region-state view
FR-5	Story	Guided narrative covering the trend shift, regional variation, and lockdown recovery scenarios
FR-6	Publishing & Web Integration	Publish to Tableau Public/Server; embed the dashboard in a webpage

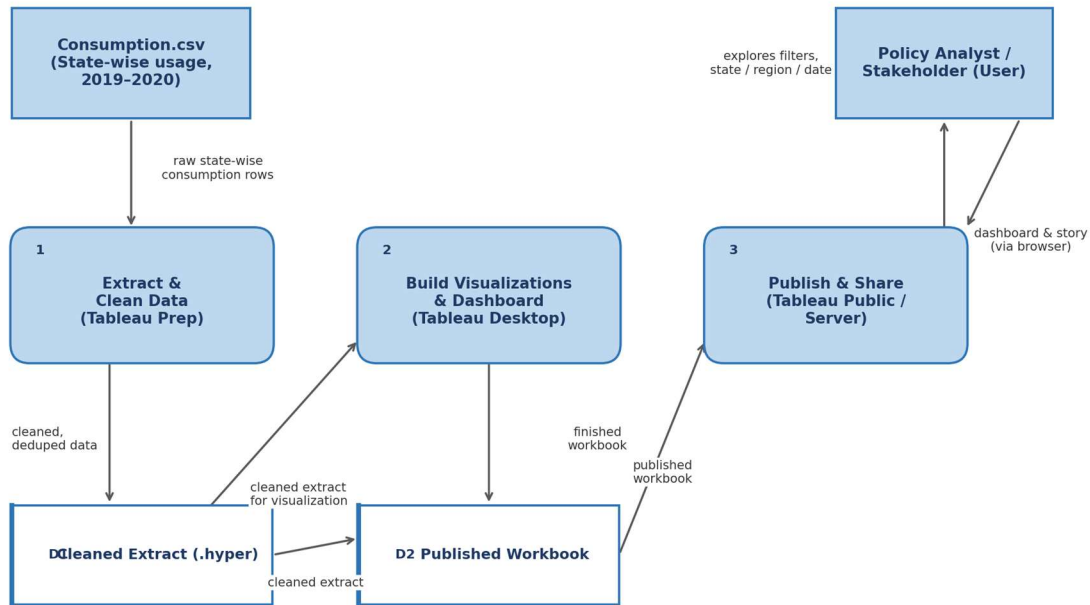
Non-Functional Requirements:

NFR No.	Requirement	Description
NFR-1	Usability	Understandable by non-technical stakeholders within minutes; no Tableau training required.
NFR-2	Performance	Filters respond within 2–3 seconds across the ~16,600-row dataset using extracts.
NFR-3	Accuracy	Calculated fields validated against manual spot-checks before publishing.
NFR-4	Accessibility	Viewable on desktop/mobile browsers via Tableau Public/Server without a license.
NFR-5	Maintainability	Clear worksheet/field naming so the workbook can be updated as new data arrives.

NFR No.	Requirement	Description
NFR-6	Scalability	Extract-based design accommodates additional years/states without a rebuild.

3.3 Data Flow Diagram

Data Flow Diagram — Level 0 (Electricity Consumption Analytics, Tableau)



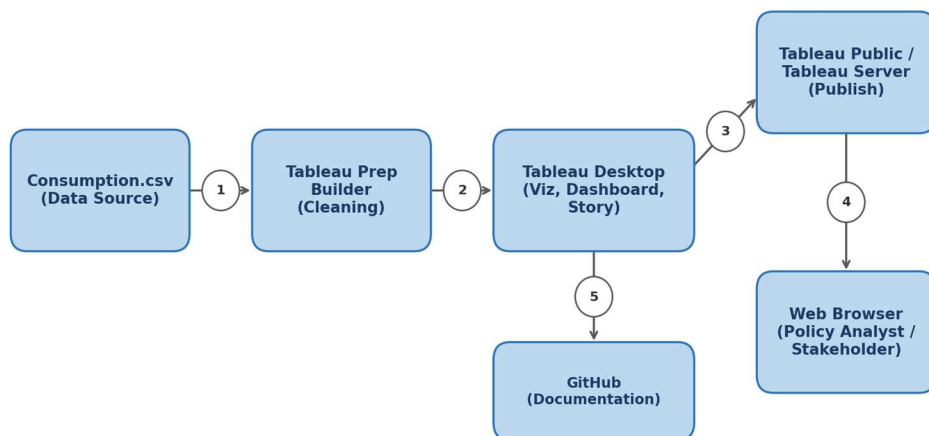
Level-0 DFD: data source → extract & clean → build dashboard → publish & share

Key user stories:

Story #	Epic	User Story / Task	Priority	Release
USN-1	Data Integration	Connect Tableau to the Consumption.csv file.	High	Sprint-1
USN-2	Data Preparation	Clean and validate data (dedupe, fix state names) in Tableau Prep.	High	Sprint-1
USN-3	Data Preparation	Create a calculated field flagging the lockdown period.	High	Sprint-1
USN-4	Data Visualization	View a line/bar chart of consumption by region and year.	High	Sprint-2
USN-5	Data Visualization	Filter by region so I can compare demand across NR/SR/ER/WR/NER.	High	Sprint-2
USN-6	Data Visualization	View a map of average state-wise consumption.	Medium	Sprint-2
USN-7	Dashboard	View a Top N / Bottom N ranking of states by usage.	High	Sprint-2
USN-8	Story	Walk through a guided Tableau Story covering the three scenarios.	High	Sprint-3
USN-9	Publishing	Publish the workbook to Tableau Public/Server.	Medium	Sprint-3
USN-10	Web Integration	Interact with dashboard filters on the published web view.	Medium	Sprint-3

3.4 Technology Stack

Technology Stack — Electricity Consumption Analytics (Tableau)



Technology stack: CSV → Tableau Prep → Tableau Desktop → Tableau Public/Server → Browser

S.No	Component	Description	Technology
1	Data Source	Historical state-wise consumption records (2019–2020)	CSV file (Consumption.csv)
2	Data Preparation	Cleaning, deduplication, field creation	Tableau Prep Builder
3	Visualization & Analysis	Building charts, dashboards, and the story	Tableau Desktop
4	Calculated Fields	Lockdown flag, Top N / Bottom N rank	Tableau calculated fields
5	Geocoding / Mapping	Plotting state-wise consumption geographically	Built-in geocoding + lat/long fields
6	Publishing Platform	Hosting the interactive dashboard and story	Tableau Public / Server
7	Web Embedding	Embedding the dashboard on a webpage	Tableau JavaScript API
8	Version Control	Tracking project files and documentation	GitHub

4. Project Design

4.1 Problem Solution Fit

Problem - Solution Fit Canvas — Electricity Consumption Analytics (Tableau)

Define CS, fit into CC Focus on J&P, understand RC Identify strong TR & EM	1. CUSTOMER SEGMENT(S) CS Energy policy analysts & grid planners analyzing 2019-2020 state consumption data; researchers needing to communicate COVID-19 impact to non-technical stakeholders.	6. CUSTOMER CONSTRAINTS CC Limited advanced analytics/BI training, no dedicated BI team access, tight time/budget constraints, heavy reliance on Excel for day-to-day reporting.	5. AVAILABLE SOLUTIONS AS Manual Excel pivot tables & charts; static PDF slide decks; ad-hoc requests to a central data/IT team for custom reports.	Explore AS, differentiate Focus on J&P, tap into BE, understand RC Extract online & offline CH of BE
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Understand the overall consumption trend shift; compare regions; assess how quickly each state recovered after the 2020 lockdown.	9. PROBLEM ROOT CAUSE RC The raw flat CSV of state-date consumption rows has no built-in trend, regional, or lockdown-recovery visualization capability.	7. BEHAVIOUR BE Analysts manually filter and pivot the data in Excel, build static charts for each report, and repeat this work whenever a new question comes up.	
	3. TRIGGERS TR A quarterly policy review; a leadership request for a COVID impact summary; the start of a new fiscal-year planning cycle.	10. YOUR SOLUTION SL An interactive Tableau dashboard and guided story built from the 2019-2020 state consumption dataset, visualizing the national trend shift, regional variation, and the pace of post-lockdown recovery — replacing manual Excel work with one interactive, shareable view.	8.1 ONLINE CH Existing Excel/PDF reports, internal email bulletins, presentation decks shared over email or Teams.	
	4. EMOTIONS: BEFORE / AFTER EM Before: overwhelmed, unsure how to explain the findings. After: confident, articulate, and backed by clear data.		8.2 OFFLINE In-person planning meetings, printed briefing documents, verbal updates in review sessions.	

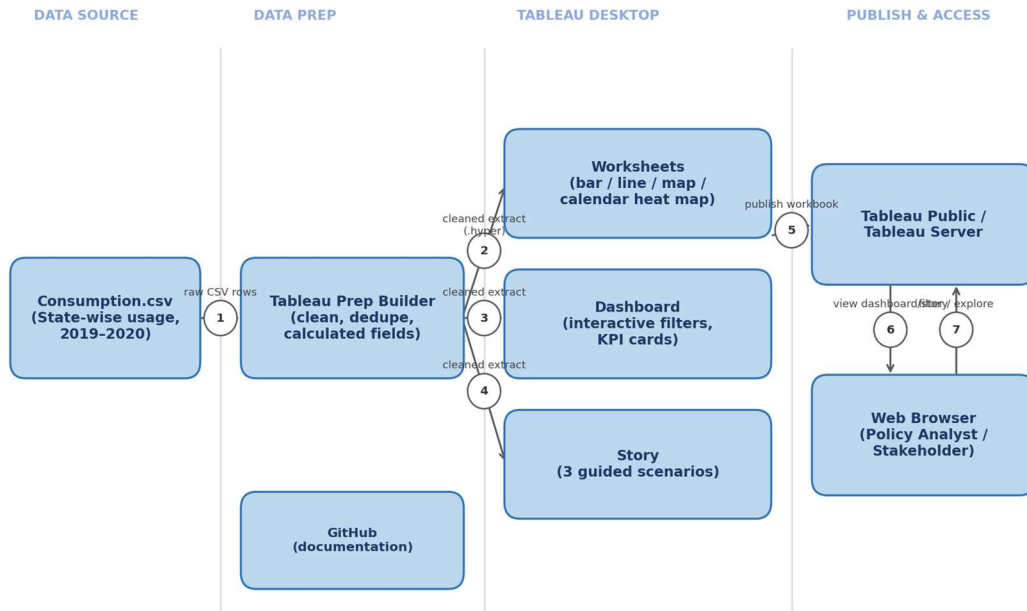
Problem–Solution Fit canvas

4.2 Proposed Solution

S.No	Parameter	Description
1.	Problem Statement	Analysts and planners lack an accessible way to see how India's state-wise electricity consumption shifted from 2019 to 2020 — including the COVID-19 lockdown's impact and the uneven pace of recovery — without hours of manual spreadsheet work.
2.	Idea / Solution	An interactive Tableau dashboard and guided story built from the 2019–2020 consumption dataset, visualizing the national trend shift, regional variation, and the pace of post-lockdown recovery.
3.	Novelty / Uniqueness	Combines the trend, regional, and recovery lenses into one connected Tableau story; a calculated lockdown-period flag makes before/during/after comparisons a single click.
4.	Social Impact	Helps policymakers make faster, better-informed energy planning decisions and gives analysts a reusable template for future crisis-impact analysis.
5.	Business Model	Delivered as an analytics consulting deliverable to state electricity boards, or published openly on Tableau Public for transparency.
6.	Scalability	The extract-based structure and calculated fields extend to new years of data or additional states/regions without rebuilding the analysis.

4.3 Solution Architecture

Solution Architecture – Electricity Consumption Analytics (Tableau)



Solution architecture: from raw CSV to published Tableau dashboard

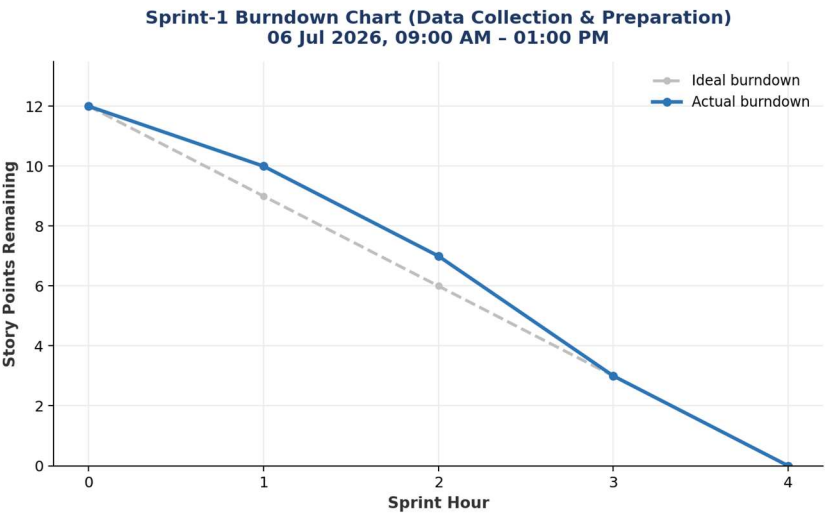
5. Project Planning & Scheduling

5.1 Project Planning

The backlog was organized into 4 sprints aligned to the project's modules, compressed to fit the submission timeline (06–07 July 2026):

Sprint	Epics Covered	Story Points (breakdown)	Total
Sprint-1	Data Collection & Extraction	2+1+3+3+3	12
Sprint-2	Data Visualization + Dashboard + Story	2+2+2+4+5+5	20
Sprint-3	Performance Testing + Web Integration	3+2+3+3+2	13
Sprint-4	Project Demonstration & Documentation	3+3+2+2	10

Velocity = Total Story Points / Number of Sprints = $(12 + 20 + 13 + 10) / 4 = 55 / 4 = 13.8$ Story Points per Sprint



Sprint-1 burndown chart

6. Functional and Performance Testing

6.1 Performance Testing

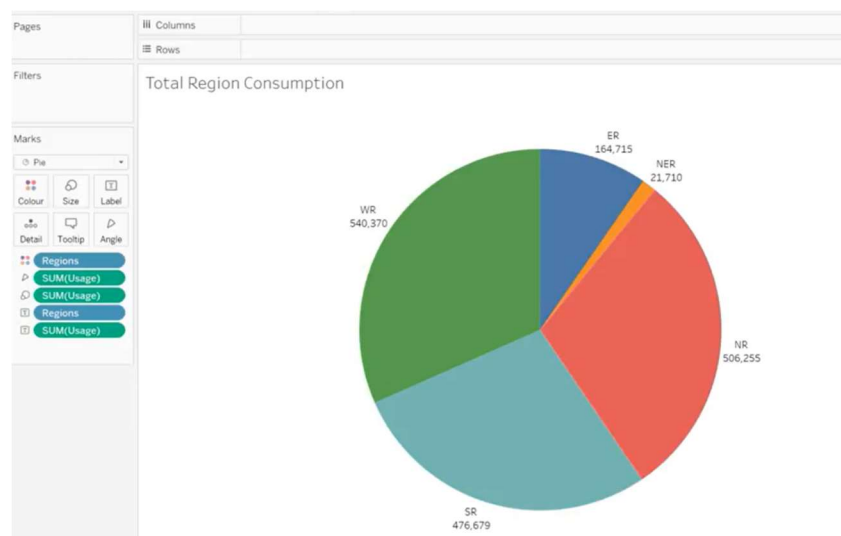
Parameter	Values
Data Rendered	Consumption.csv rendered across 4 worksheets: Usage by Region, 2019 State Consumption, 2020 State Consumption, Total Consumption.
Data Preprocessing	Standardized state names; confirmed no missing values; converted Date to date type; accounted for the reduced reporting window in 2020.
Utilization of Filters	YEAR(Dates) filter (2019/2020); Regions filter/legend; States filter driving Top N / Bottom N; Top N and Bottom N parameters (set to 5).
Calculation Fields Used	Top N Parameter; Top N Rank = RANK(SUM([Usage]), 'desc'); Bottom N Rank = RANK(SUM([Usage]), 'asc').

7. Results

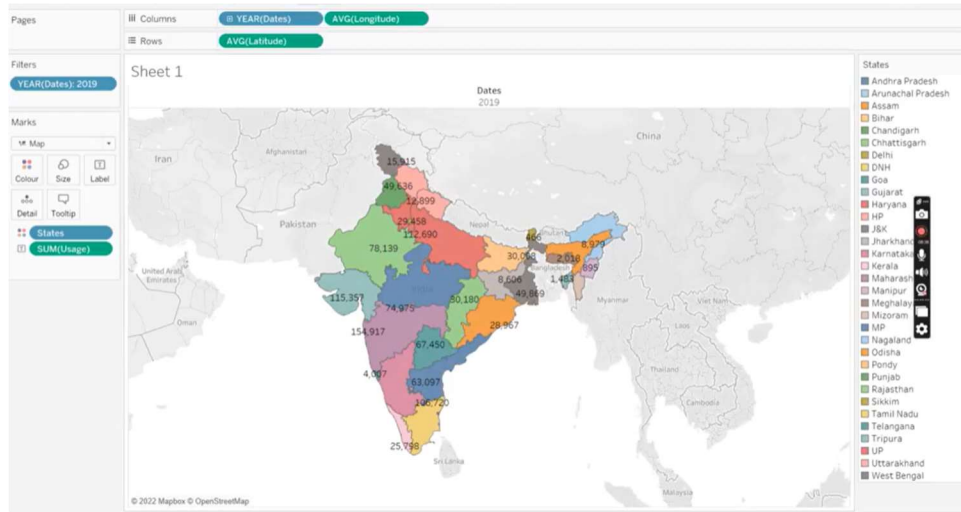
7.1 Output Screenshots



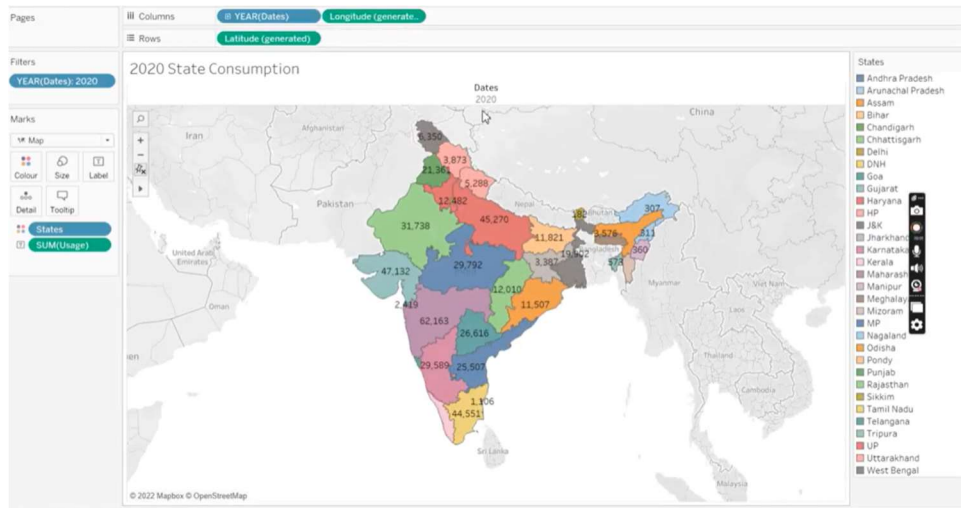
Usage by Region — bar chart split by year (2019 vs 2020)



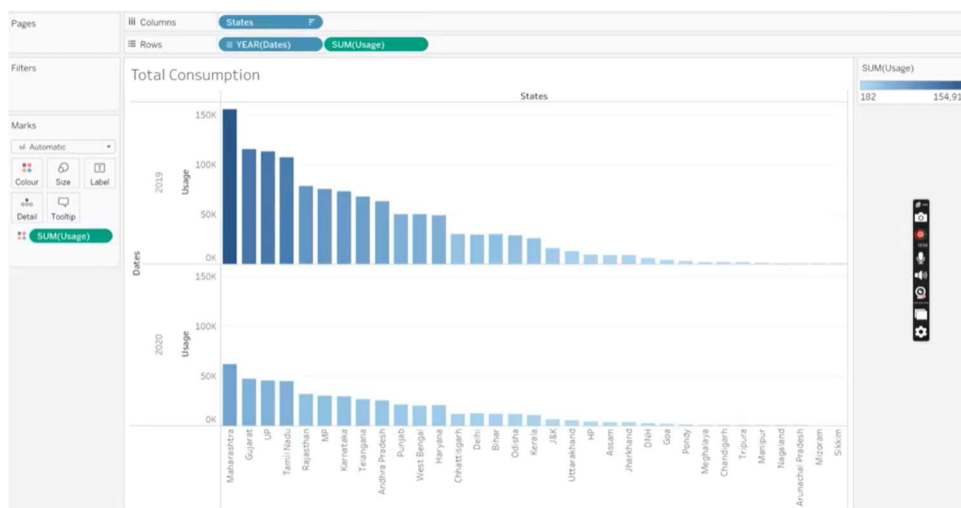
Total Region Consumption — pie chart



2019 State Consumption map



2020 State Consumption map



Total Consumption by state, 2019 vs 2020



Region Wise State Usage dashboard with Top N / Bottom N rankings

8. Advantages & Disadvantages

Advantages

- Consolidates three years of scattered daily records into a single, interactive view of national, regional, and state-level trends.
- No coding required for stakeholders — accessible through any browser once published.
- The calculated lockdown-period flag and Top N/Bottom N rankings make before/after and best/worst comparisons a single click.
- Reusable as a template for future crisis-impact or year-over-year analyses.
- Region and state-level granularity supports targeted, localized policy decisions.

Disadvantages

- The sharp drop in reported rows per state in 2020 limits direct comparison of raw yearly totals.
- Tableau Public dashboards are publicly visible by default; private hosting requires Tableau Server, which has a licensing cost.
- The extract-based data source requires a manual refresh whenever new data becomes available — there is no live/real-time feed.
- The project is limited to historical, descriptive analysis; it does not include predictive forecasting.

9. Conclusion

This project demonstrates how a large, unstructured dataset of state-wise electricity consumption can be transformed into a clear, interactive analysis using Tableau. By focusing on three concrete scenarios — the overall trend shift, regional variation, and lockdown recovery — the resulting dashboard and story give energy analysts and policymakers a tool to explore India's 2019–2020 consumption patterns without needing to process raw spreadsheets themselves. The workflow, from data cleaning through calculated fields to a published, interactive dashboard, reflects a complete and practical application of the data analytics lifecycle.

10. Future Scope

- Extend the dataset with more recent years to track longer-term recovery and growth trends.
- Integrate a live or scheduled data feed so the dashboard updates automatically rather than through manual extract refreshes.
- Add a forecasting layer (e.g., using Tableau's built-in forecasting or an external model) to project future demand.
- Host the dashboard on Tableau Server with role-based access control for state electricity boards that require private data.
- Build a lightweight mobile-friendly view or embed the dashboard directly into state energy department websites.

11. Appendix

Source Code (if any)

This project is primarily a Tableau workbook (data preparation, calculated fields, dashboard, and story); no separate application backend was built. Data preparation logic is documented in Sections 3.2 and 6.1.

Dataset Link

Consumption.csv — state-wise daily electricity consumption data for India, January 2019 to December 2020 (33 states, 5 regions), provided as part of the project materials.

GitHub & Project Demo Link

GitHub Repository: <https://github.com/Arjunatwar/Plugging-into-the-Future/tree/main>

Project Demo: <https://drive.google.com/drive/folders/1WomE8CdxmuzajJlKV8Frgk97AvVy7eWc?usp=sharing>